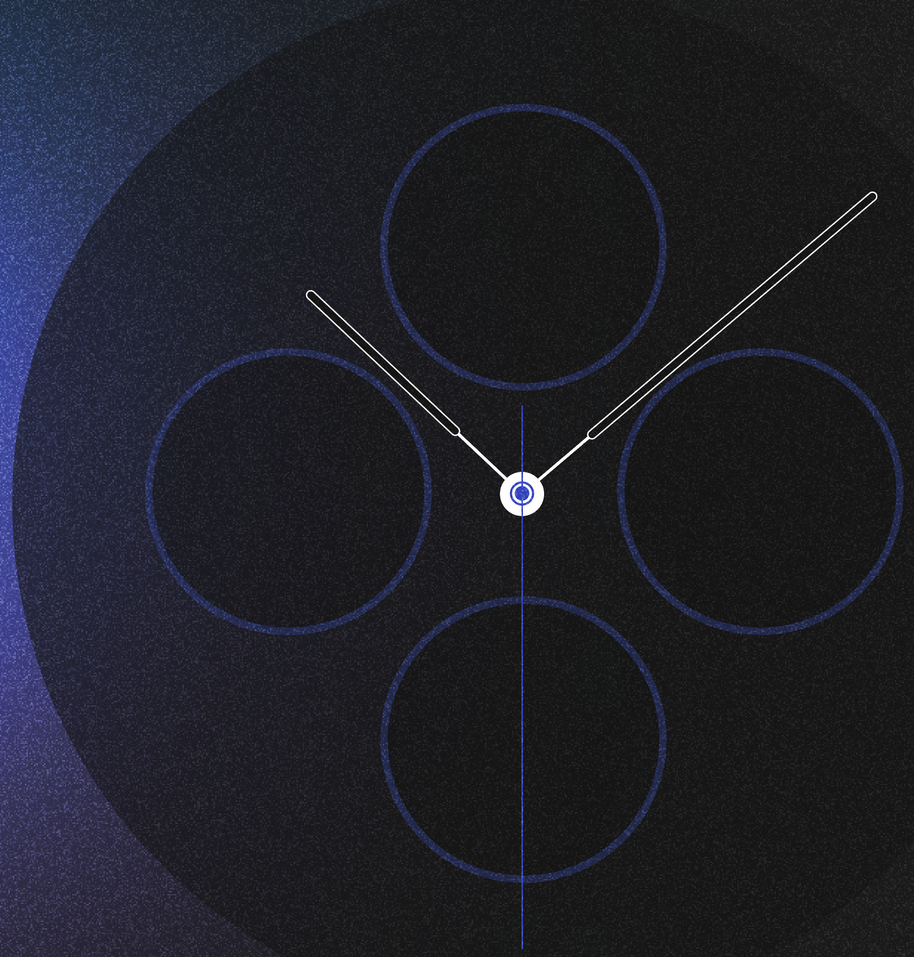
Neurodivergence and its impact on time perception

VR Capstone 2023 | Team Fantastic Four

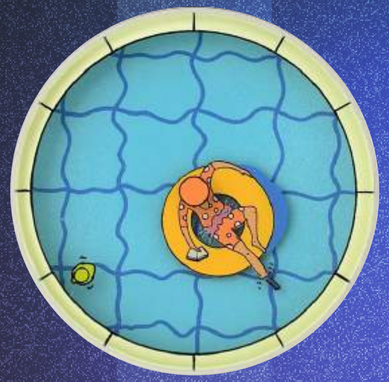


The Team
Executive Summary
Project Background
Research Methodology
Ideation & Prototyping
Testing & Iteration
Design Requirements & Principles
Final Product
Limitations & Challenges
Future Work
Reflection
Appendix



Haasini Sai

Researcher and product designer specialized in product design and story boarding



Yifan Lin

Researcher & UX designer focusing on data analysis and accessibility



Parag Nandi

UX designer merging human interaction & tech, with focus on privacy and accessibility



Gresshaa Mehta

Product designer & researcher specializing in interaction design & accessibility.

Executive Summary

In the Fall of 2022, our team saw a series of videos of people on TikTok sketching how they visualize calendars. We noticed that each person in the videos visualized calendars in a different way - circular, mountain peaks, spirals and more. We dove deeper into this phenomenon and found that people who can visualize calendars spatially have time-space synesthesia (TSS). People with TSS have a cognitive advantage over people who cannot visualize time spatially or visually.

We noticed that adults with Autism and/or ADHD in particular struggled with time perception, blindness, and horizon. Our primary hypothesis was questioning whether we could apply visualization techniques from TSS to help people who struggle with time perception.

From our diary studies, user interviews, and co-design sessions with our participants in 2023, we found that people associate time with change but cannot notice those changes when immersed in a task. We prototyped a VR game that trains people to notice time related changes in the environment around them.

This document provides insight into our capstone project from the 2023 academic year. We highlight our project background, goals, key definitions, key stakeholders, research methodology, design process, limitations, principles, requirements, and more. We spent this quarter conducting co-design sessions, prototyping, and usability testing. We critically evaluated the effectiveness of our prototype and dove deep into the potential dangers of our work. We'd like to thank Professor Elin Björling for their support throughout our messy design process.

Background



Time Perception

The ability to perceive, understand, and use time. It is measured by ones own perception of the duration of the indefinite and unfolding of events

Time Blindness

The inability to sense the passing of time. It’s a sensory issue, not an intentional disregard for time.

Time Horizon

The period of time that we can work within. It is the point in the future when things stop feeling real and begin to feel imaginary.

Project Inception

Found a series of TikToks in Fall 2022 of people drawing calendars and describing how they can “touch”, “feel”, and “stand inside” time.



Circular Calendar



Mountain Calendar



Bubble Calendar

Background

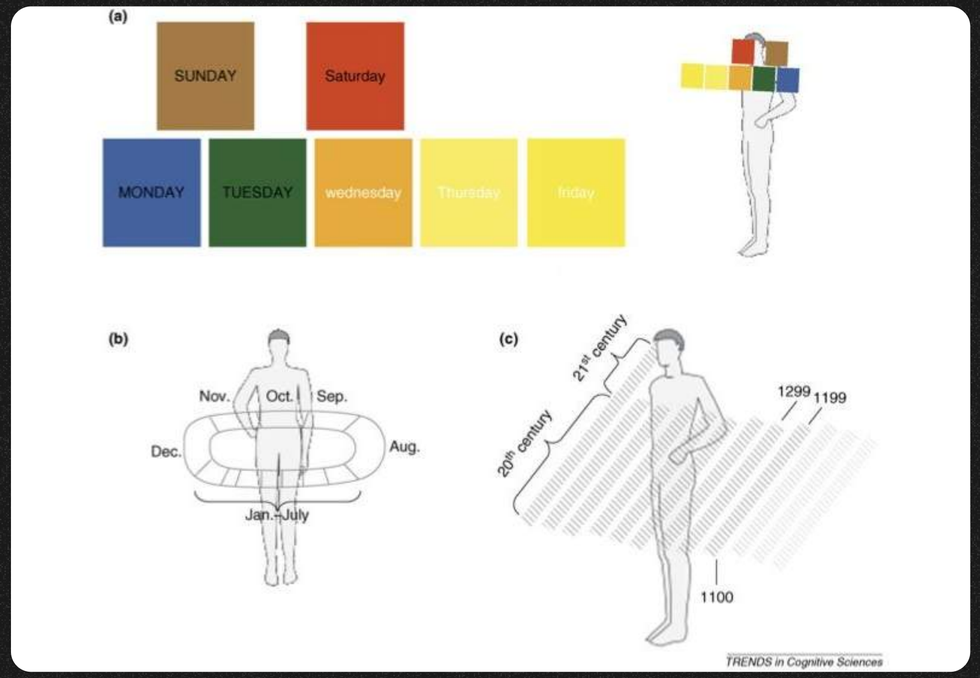
Autism affects one out of 59 people and ADHD affects around 8 million adults in the US. Time perception is a cluster of cognitive processes such as time blindness and time horizon that neurodivergent adults struggle with

Problem

How might we help adults with autism and/or ADHD improve their perception of time using immersion tools?

Target User

Working neurodivergent adults (28-40) in the USA that struggle with time perception, specifically those who have Autism and/or ADHD



People on the spectrum who could visualize time spatially

How did I get so late so soon? A review of time processing and management in autism

Lucie Jurek [1], Yannick Longuet [2], Matias Baltazar [3], Anouck Amestoy [4], Vicky Schmitt [5], Michel Desmurget [6], Marie-Maude Geoffray [7]

Affiliations + expand

PMID: 31376445 DOI: 10.1016/j.bbr.2019.112121

Abstract

While the definition of
concerning difficulties
temporal perception ir
current literature on ti
assessment tools, we
infra-second level, an
levels of time process

Time Perception is a Focal Symptom of Attention-Deficit/Hyperactivity Disorder in Adults

Simon Weissenberger,[A,E,F] Katerina Schonova,[E,F] Pascal Büttiker,[E,F] Raffaele Fazio,[E,F] Martina Vnukova,[E,F] George B. Stefano,[A,E] and Radek Ptacek[✉A,E,G]

Author information · Article notes · Copyright and License information Disclaimer

Abstract Go to:

Attention-Deficit/Hyperactivity Disorder (ADHD) is classically associated with symptoms that include inattentiveness, hyperactivity, and impulsivity together with a variety of other observable externalized symptoms. ADHD has also been associated with specific internalized cognitive symptoms, including restlessness and emotional impulsivity. This disorder has been recognized as a lifelong condition and can be recognized by a variety of unique cognitive phenomena. In addition to the frequently ignored affective symptoms exhibited by individuals diagnosed with ADHD, problems with time perception have been noted, although these are considered to be secondary issues. Temporal shifts in cognitive processing, however, may be at the very root of ADHD-related symptoms, given the importance of coordinated signal

Our focus: People on the spectrum who have trouble grasping the concept of time

Research



How might we help adults* with autism and/or ADHD improve their perception of time using immersion tools?

*working adults between 28-40 in USA

Literature

Over 15 research papers, and multiple articles related to time synesthesia, VR as a training tool, neurodivergence and it's impact on time perception etc.

Interviews

60 minute interviews with five neurodivergent adults to understand nuances related to time estimation

Diary Studies

3 day diary studies with five adults to understand nuances related to day-day time estimation and perception.

Co Design

Power of threes: three sessions over three weeks with three separate neurodivergent adults in the US.

Research Goals

Understanding how people who have ADHD and/or Autism struggle with time perception.	Understanding the methods users follow in estimating time, time horizons, and reducing time blindness.	Understanding the level of comfort users have with mixed reality and leveraging them to improve time blindness

Research Questions

What challenges do people who have trouble with time perception face in their day to day life? What are the differences in which neurotypical and neurodivergent people perceive time? Can our solution help both audiences?	What are the digital and/or analog tools do people with time blindness use to keep track of time currently?	What's the difference between visualizing time in 2D vs 3D? Which one is more effective? How do people perceive time and its passage in VR? Does VR warp your perception of time?

“I’m knitting a passion project
and suddenly it’s two hours later,
it feels like I have time travelled”

“The past is still present. I can
remember things in such explicit
detail, I don’t feel that I’m removed
from earlier stages in my life.”

“It feels like I’m either too late,
too early, or constantly
checking the time.”

**“Time has passed
but I just didn’t feel it.”**

“I understand the math of time
but to be able to use time and
then function according to time
is a whole other ballgame.”

“I’m so immersed in what I am doing,
it feels like a super power (unless I
have somewhere to be in an hour)”

“I can’t reach time,
I can’t touch time.

It doesn’t seem like a
tangible item that exists. ”

Interview Key Insights

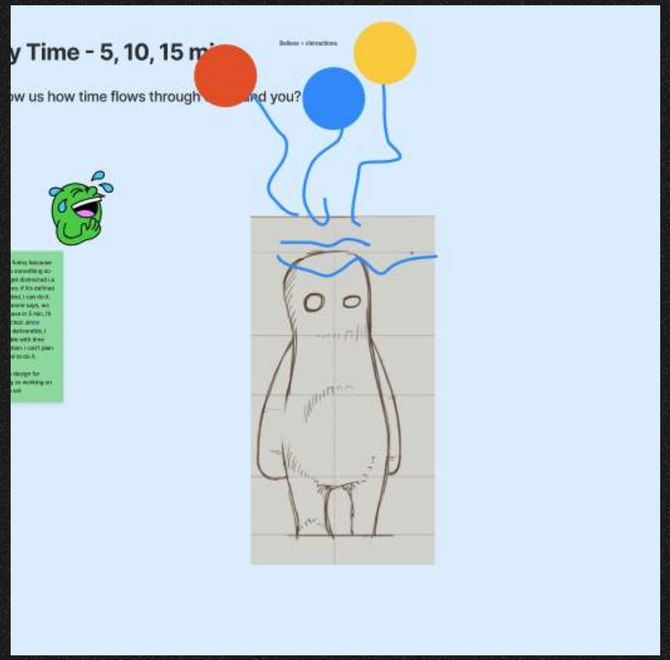
Frustrated with time estimation	75% participants found difficulty in estimating time required for completing tasks.
Unstructured Time is Precious	Participants found unstructured time to be refreshing and creatively stimulating.
Externalizing to reduce anxiety	Externalizing to-dos onto physical and digital media takes the anxiety out.
Avoiding distractions is taxing	Participants expressed struggles to avoid distraction within their scheduled timeframe.
Incentives for completing milestones	40% of participants incentivize themselves with relaxing activities for completing tasks on time

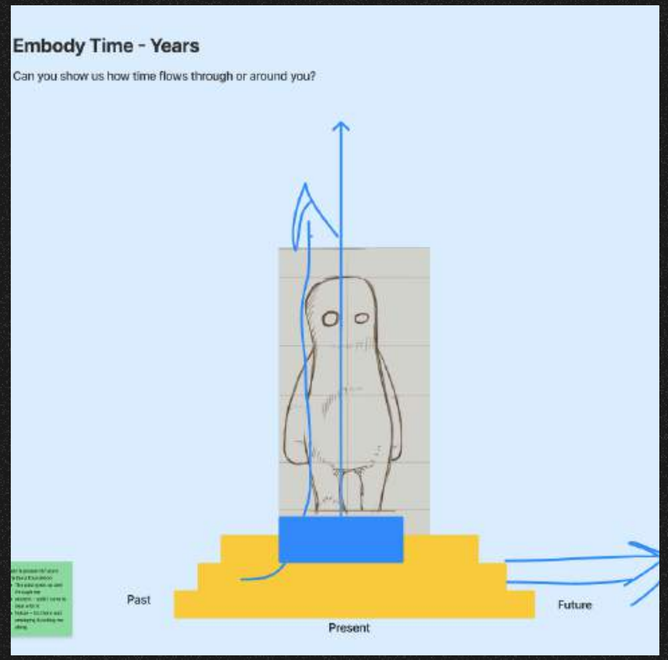
Diary Study Key Insights

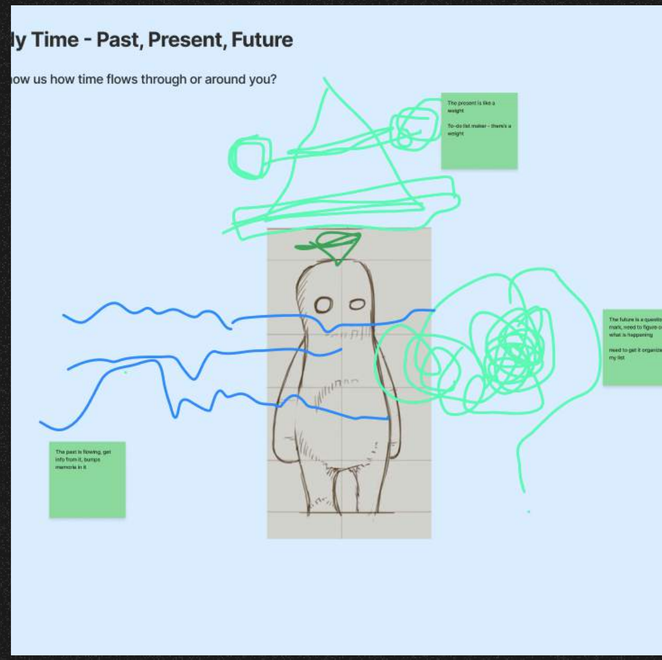
Underestimation is common	Underestimation of time happens almost everyday for all participants
Not “Productive” & no control	Feeling overwhelmed or frustrated about not being “productive” & losing control of time
Frequent usage of time blocks	80% of participants use **time blocks** to track time, understand time for a task, get reminders
Need for better estimating	Participants want to better estimate time, scope out time, structure the day
More self-care & structured time	People want themselves to have more attention to self-care time and try methods to help structuring time better (e.g. Pomodoro)

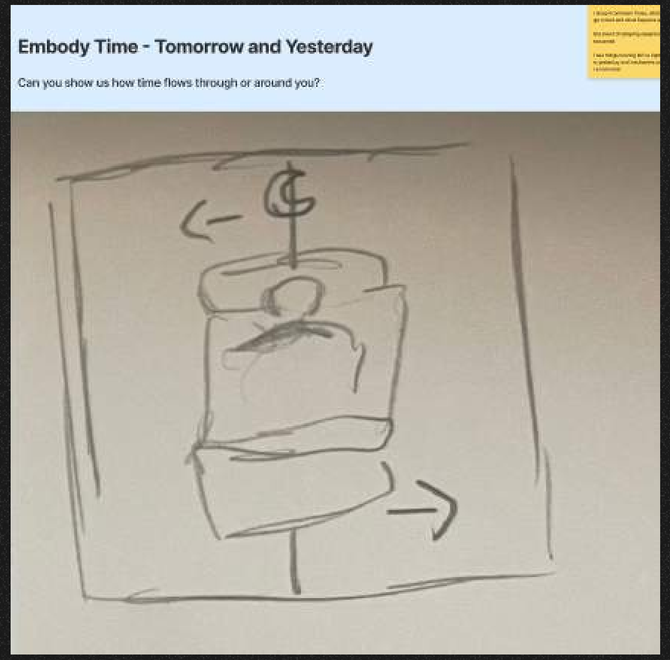
CoDesign Snippet

How does time flow around you?

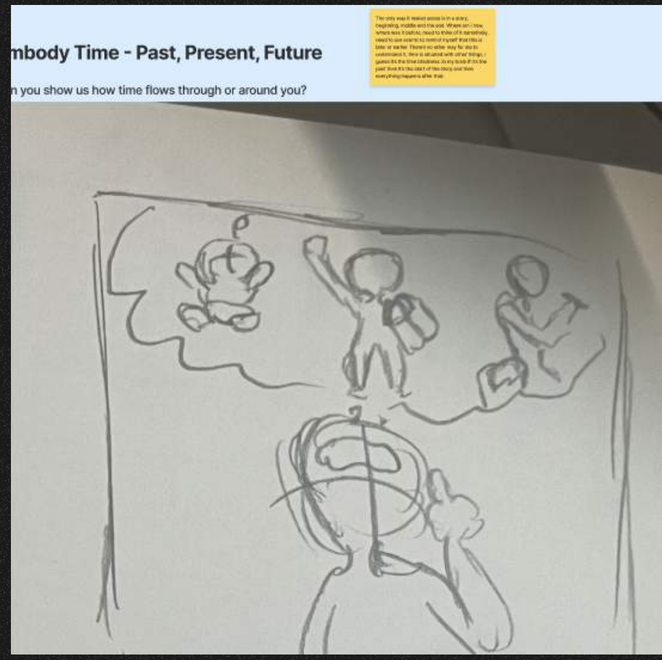












CoDesign Snippet

How do you experience time using your 5 senses?

Now



- Sunrise to sunset
- Weather
- Clocks

the sun rise and sun set from the second floor of my house. I can tell the seasons and estimate the time of day based on when i need to put the blackout shades down

kjhwilliams7530



- Chores
- Bells
- Alarm clock

whats' going on outside the house tell me what time it is- freeway traffic as an alarm clock in LA, people leaving the office at the end of the day- quieter, snoring at the same time

kjhwilliams7530



- Cooking
- temperature
- physical objects like plants

when I am making something- like clothing, a sweater or a loaf of bread- texture, weight,heft change with progress and stage. with children they feel different as they grow- not just weight but their hair texture

kjhwilliams7530



- seasons:
  - spring - flowers
  - summer - wildfires
  - fall - ?
  - winter - mint

with cooking knowing how the smell changes in different stages of the process ; Also being to smell the sun on my family when they have been out (e.g. bicycles and gardens and time in the sun)

kjhwilliams7530



- cooking: raw vs cooked meal
- perfect vs smoked
- under vs overcooked

tasting food helps me both with when the recipe is right (I have done all the work) as well as when too much time has passed

also tasting the past through memories and eating food in different parts of the country

kjhwilliams7530

Magic Wand

see the rainbow or sun in other situations, not just in my bedroom or the shower

bad at directions, can't tell b/w left and right or the cardinal directions based on where the sun is

I think I would like to hear more natural sounds vs.manmade sounds. eg waves vs. garbage truck

kjhwilliams7530

If I had a magic wand I would want to be able to summon up the touch I wanted to feel without having to spend the time hunting it down.

kjhwilliams7530

daughter remembering grandmas shirt feel

to bring memories back to life particularly with those no longer with us

kjhwilliams7530

I would like to be able to summon up the version of something that I remembered so I could enjoy it again. When companies modify recipes I can often tell and it may ruin it.

kjhwilliams7530

- Colors

- favorite songs
- sound intensity

-textures:
mornings: soft
afternoon: not soft
evening: not soft
night: squishy

fresh v. rotten

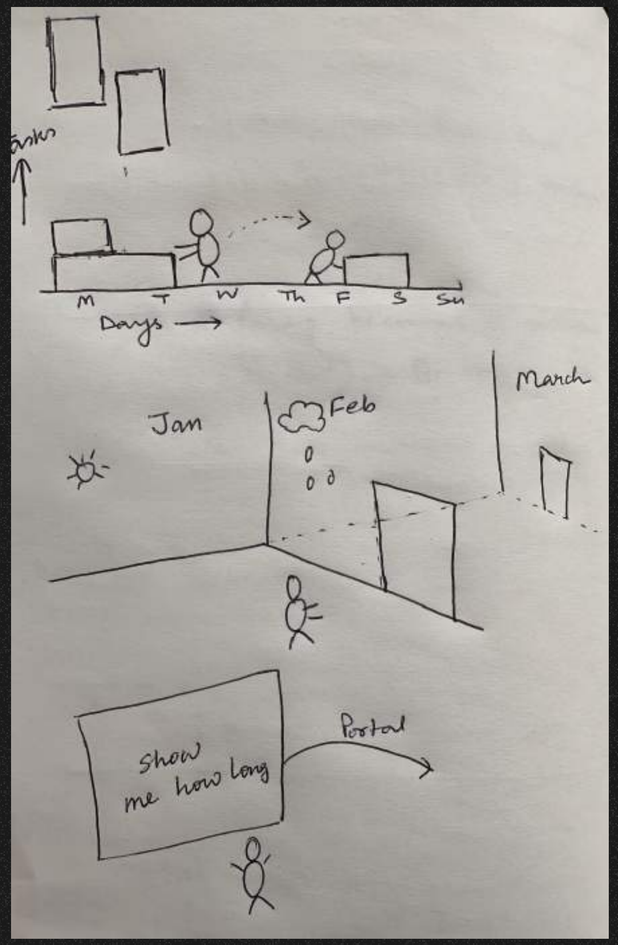
- undercooked - past
- perfect - future
- overcooked - scenario based
- cooking and tasting - present

Co-Design Key Insights

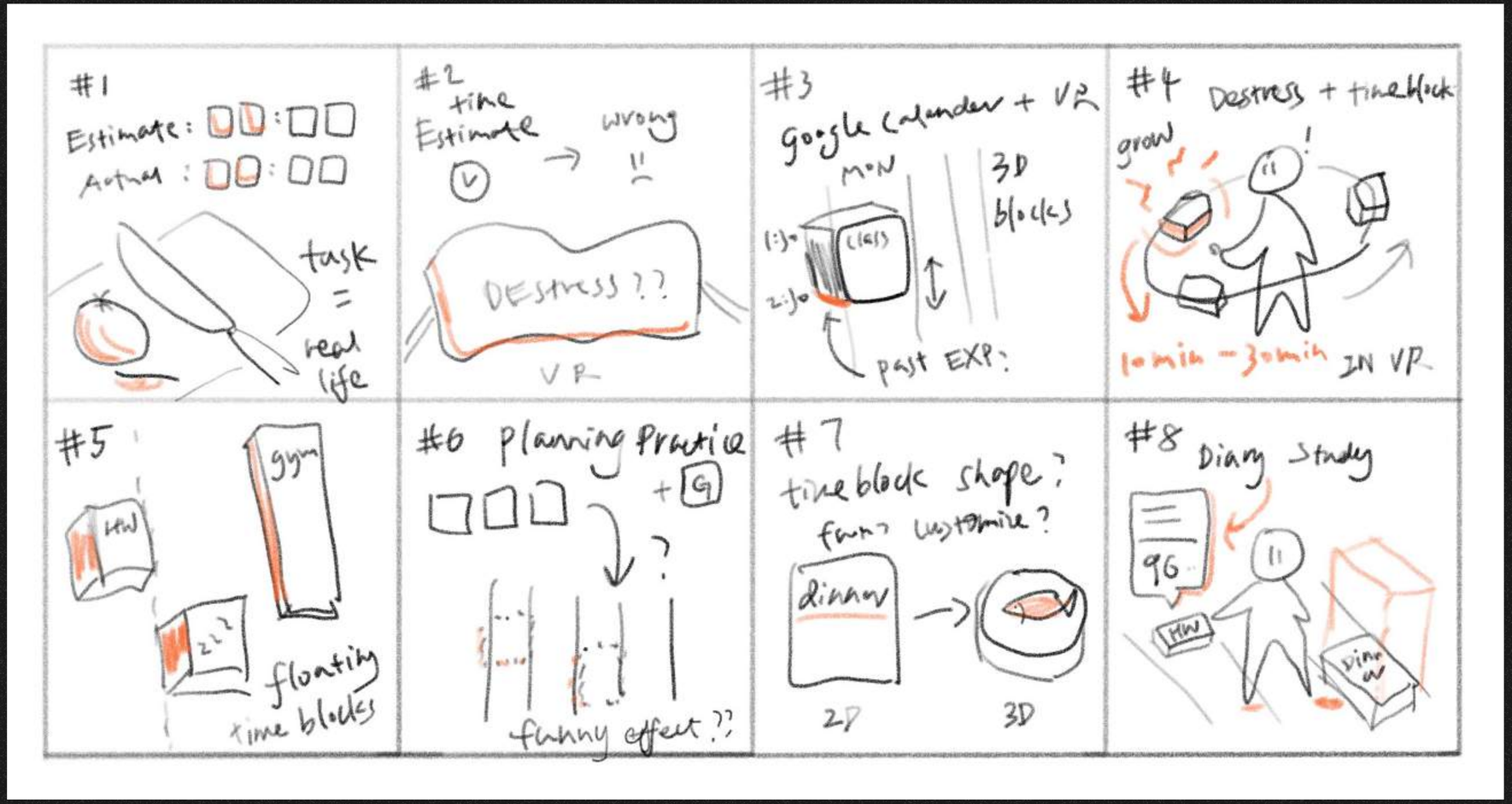
Time is observed with change	People experience time by observing the environment changing around them.
Objects make time tangible	Use physical elements like the sun, food, or the weather to feel time passing.
Time is visualized using stressors	People visualize time using stressful elements such as alarms, life or death, fresh or rotten.
Blinders up when immersed	When immersed in a task users have trouble noticing time related environmental changes.
Time is multi sensory and diverse	Each participant had a unique way of perceiving time using all their five senses

Ideation & Prototyping

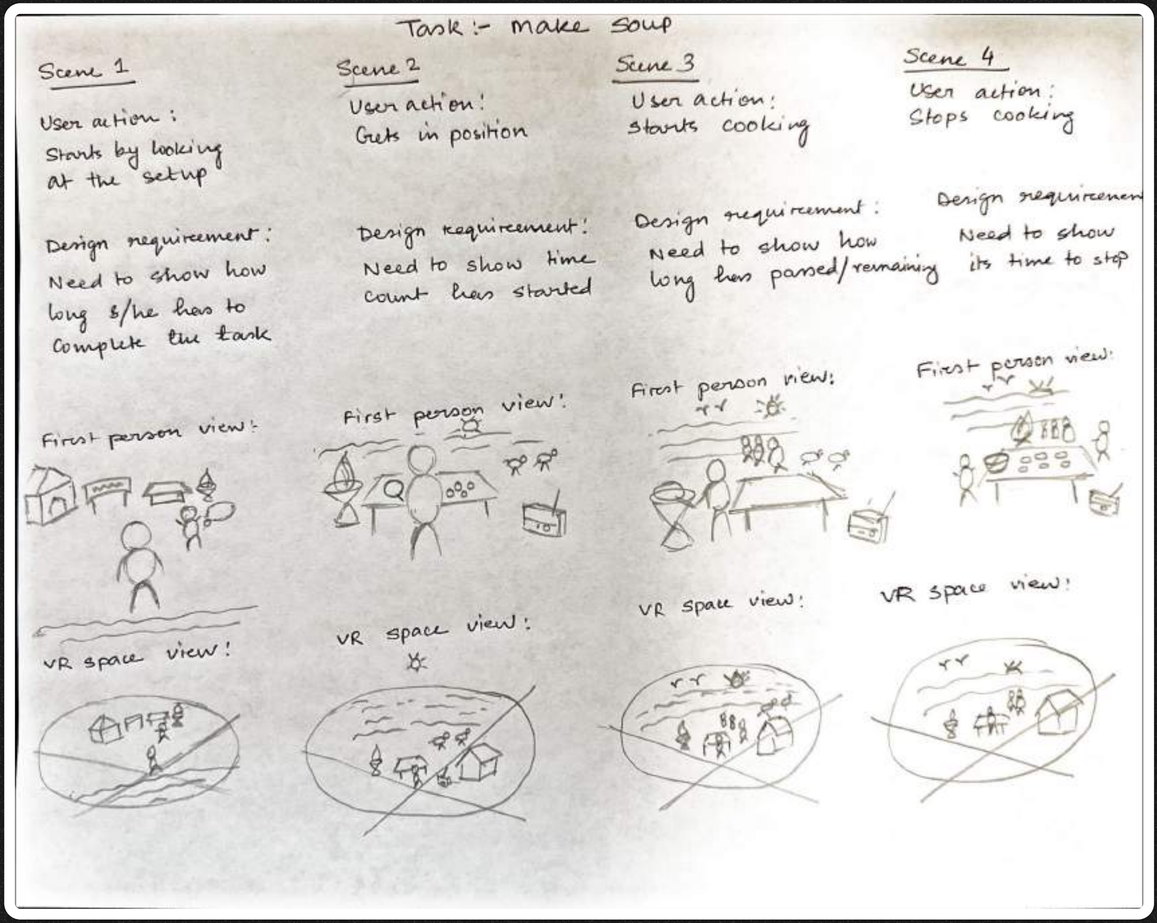


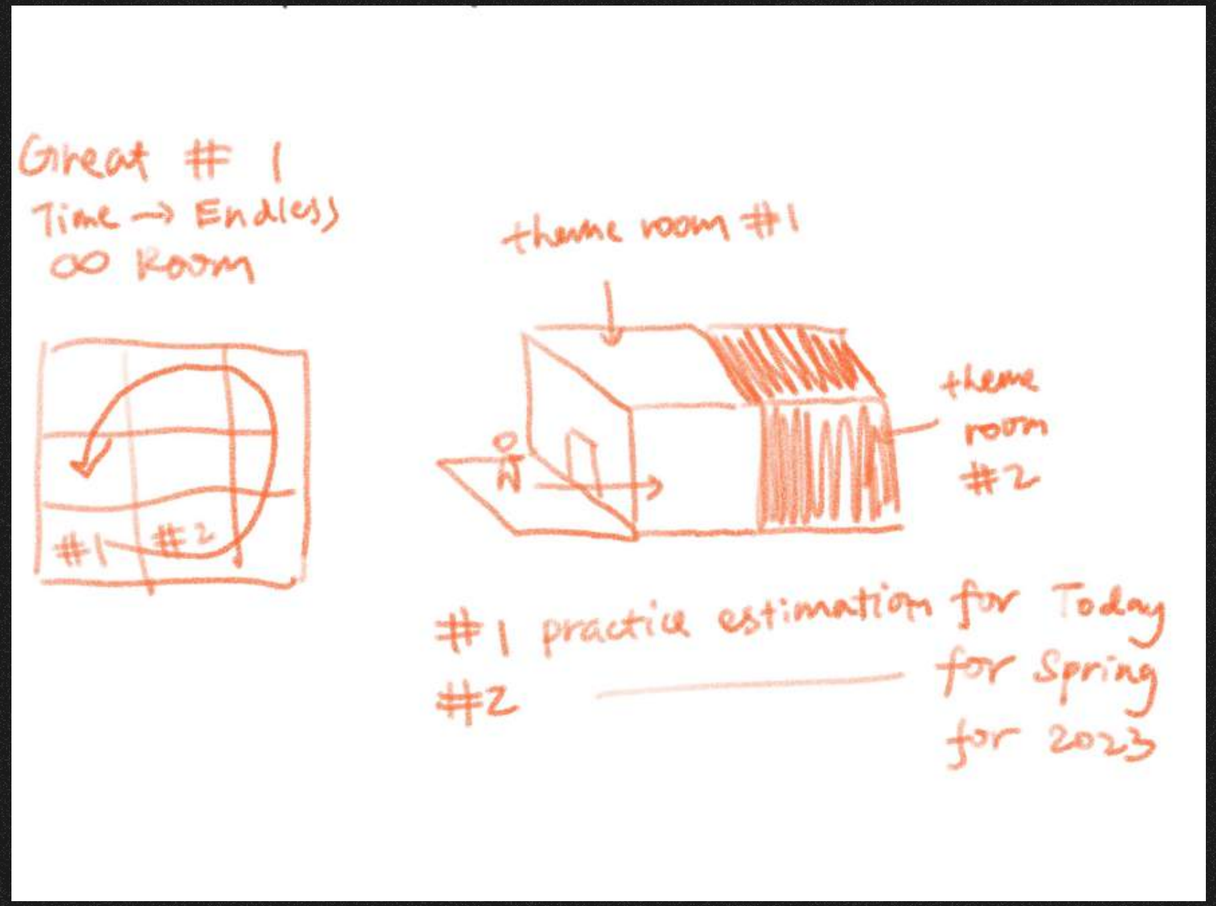


Crazy 3's Ideation Activity

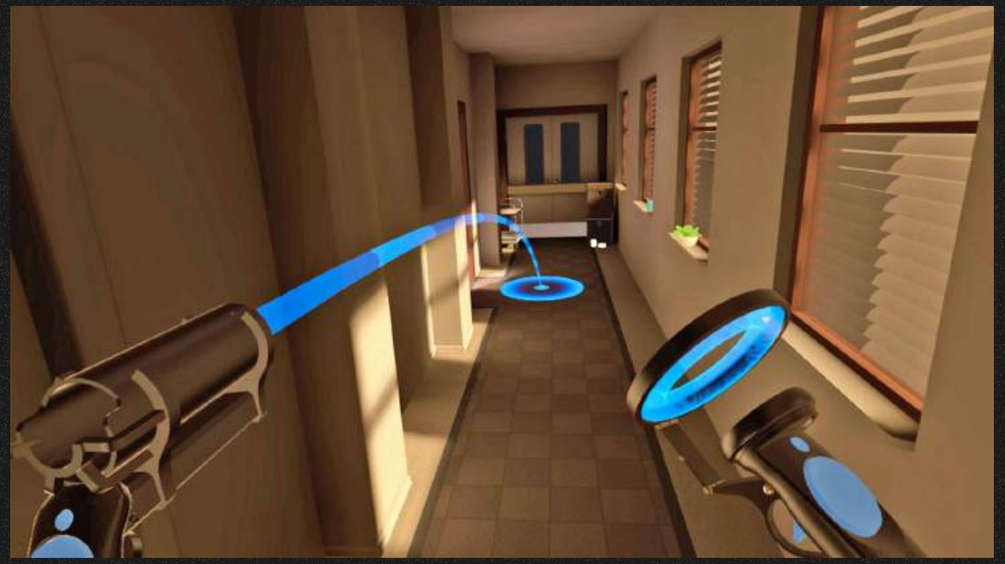


Crazy 8's Ideation Activity





Initial Storyboarding

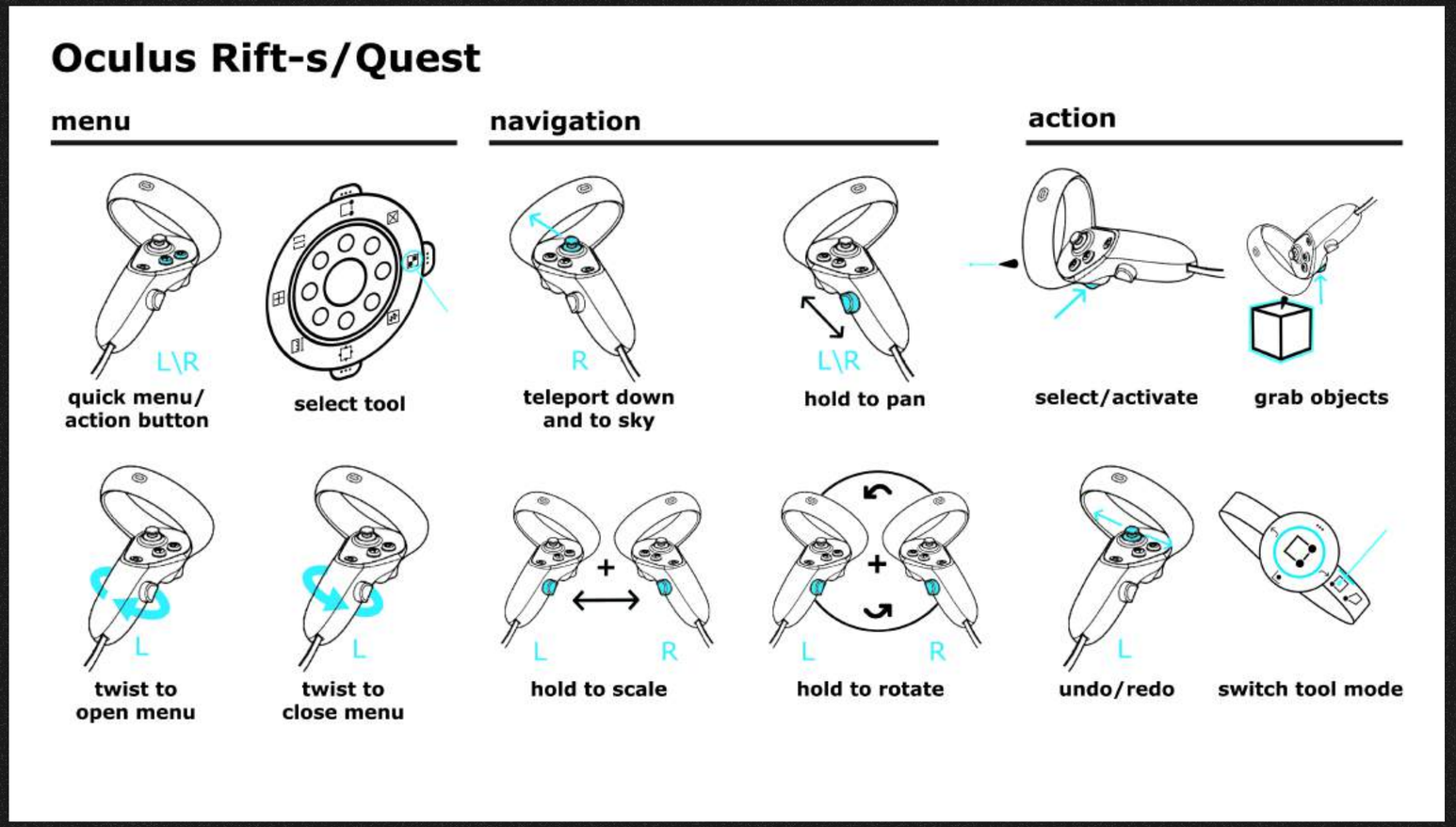


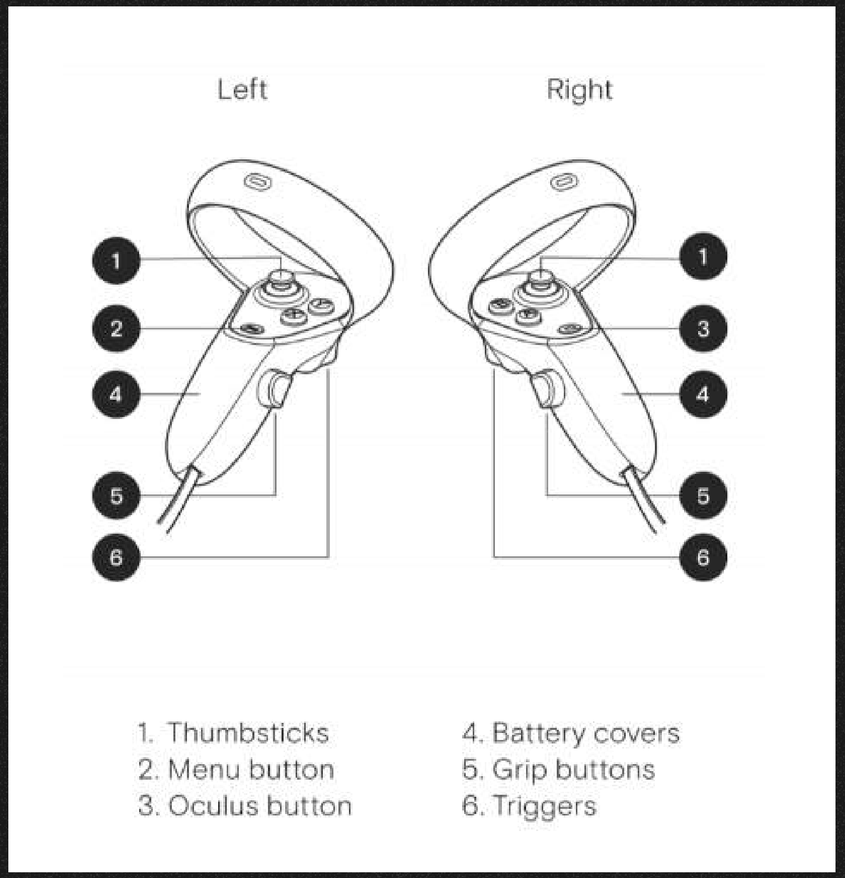






Learning from existing tools and games





Understanding and leveraging controls available to us

Idea Evolution

1.

Allow users to stand “inside” time. Create a VR calendar where people can physically move time blocks and shape them based on their needs.

2.

Create a room based environment where each room represents a moment in time eg: months, years and allow users to walk through and visualize the invisible elements of time

3.

Create an open-world VR training tool to help people notice time related changes in their environment while immersed in a task.

Hypothesis

Since users associate time with change and can't notice changes in their environment while immersed in a task, create a tool that helps them become more mindful of their surroundings while successfully completing the task in front of them.

Design Idea

Create an open-world VR training tool to help people notice time related changes in their environment while immersed in a task.

Design requirements

Tasks should be

- Individual
- Engaging & Calming
- Achievable but not boring
- Nonjudgemental measurement

Task cues should be

- Informative about time & task
- Obvious and intentional
- Only in task area
- Relatively realistic

Environment cues should be

- Calming, not stressful
- Progressively challenging
- 360 degrees
- Not invasive to task



Prototype #1: Participant working on identifying environment cues



Prototype #1: Participant working on a task



Time related environment changes eg: sun moving



Informational cues to inform the user

Testing & Iterating



Testing Questions

Environment

- How does the world make them feel?
- What are they noticing first?
- Where would they want to go?

Task

- Can they understand the task?
- Do they want to do it?
- Can they complete actions?

Time Cues

- Can they notice them?
- How do they make them feel?
- Are they triggering or relaxing?

Testing Insights

Users wanted to mimic reality	Users were confused by the dog talking, missing steps in cooking, lack of a beach and more.
Non-invasive guides and cues	Users wanted navigational guides in the environment that weren't invasive.
Sensory Overload	Some of the sound related cues eg: crickets at night were overly stimulating for users
Triggering animals & objects	Users were triggered by some of the animals and objects we had in the environment
Asking too much of users	Asking users to complete a task AND notice environmental cues was very overwhelming.

Evolution of prototype

Before		After
Time related cue identification while working on a task	→	Level based system: work your way up from noticing cues to being able to do so while working on a task
Verbal guides for navigating the environment	→	Both verbal & non verbal guides for navigating the environment
Triggering animals and objects	→	Removed object that looked like a church and removed stressful animals
Mix of reality and fantasy	→	Mimicking reality as closely as possible



Before: Cue identification while working on a task



After: Level based system where you work your way up from noticing cues to being able to do so while working on a task

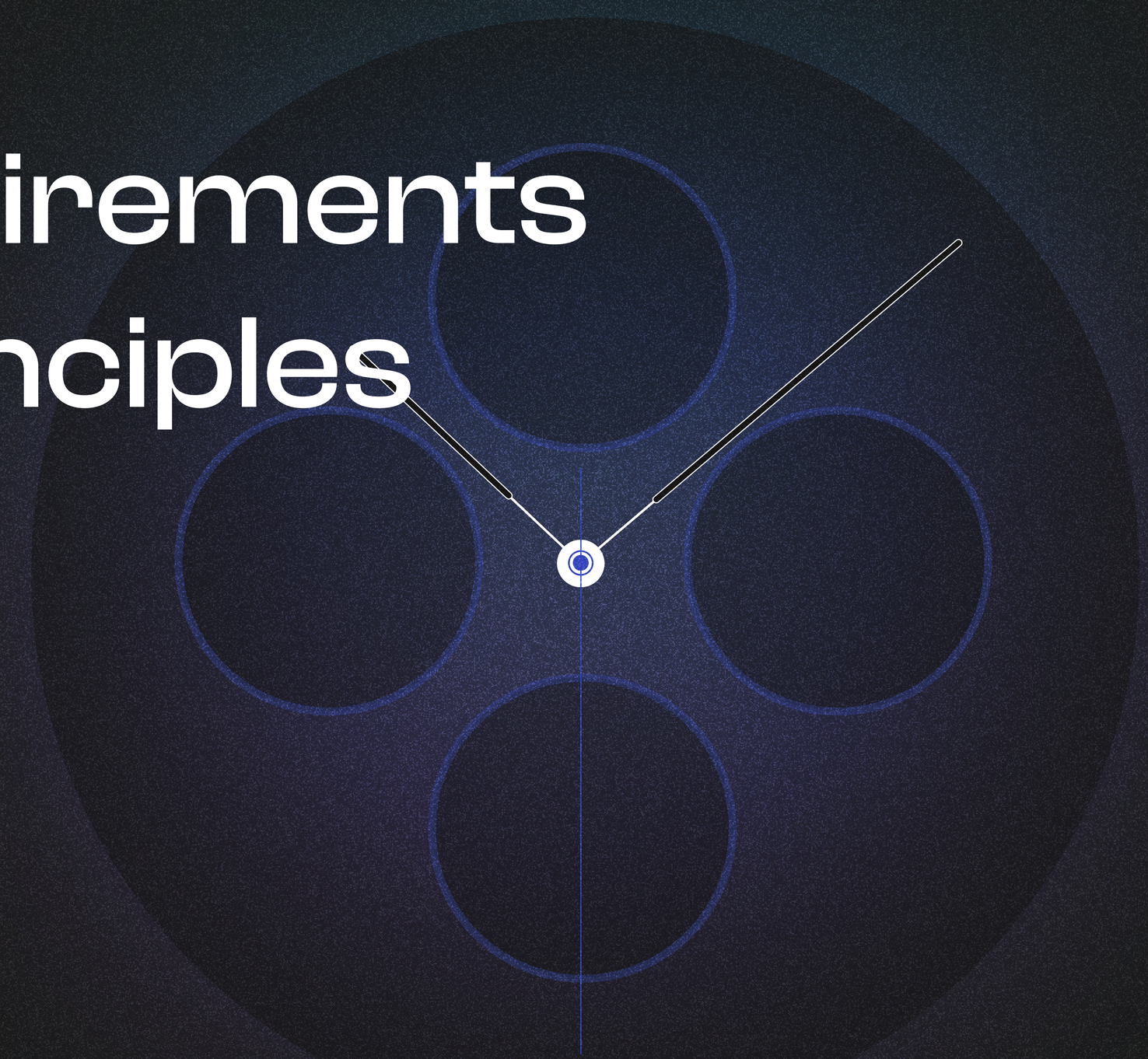


Before: Crabs out in the day, on land



After: Crabs out on a sandy beach at night (participants wanted our environment to mimic reality as closely as possible)

Requirements & Principles



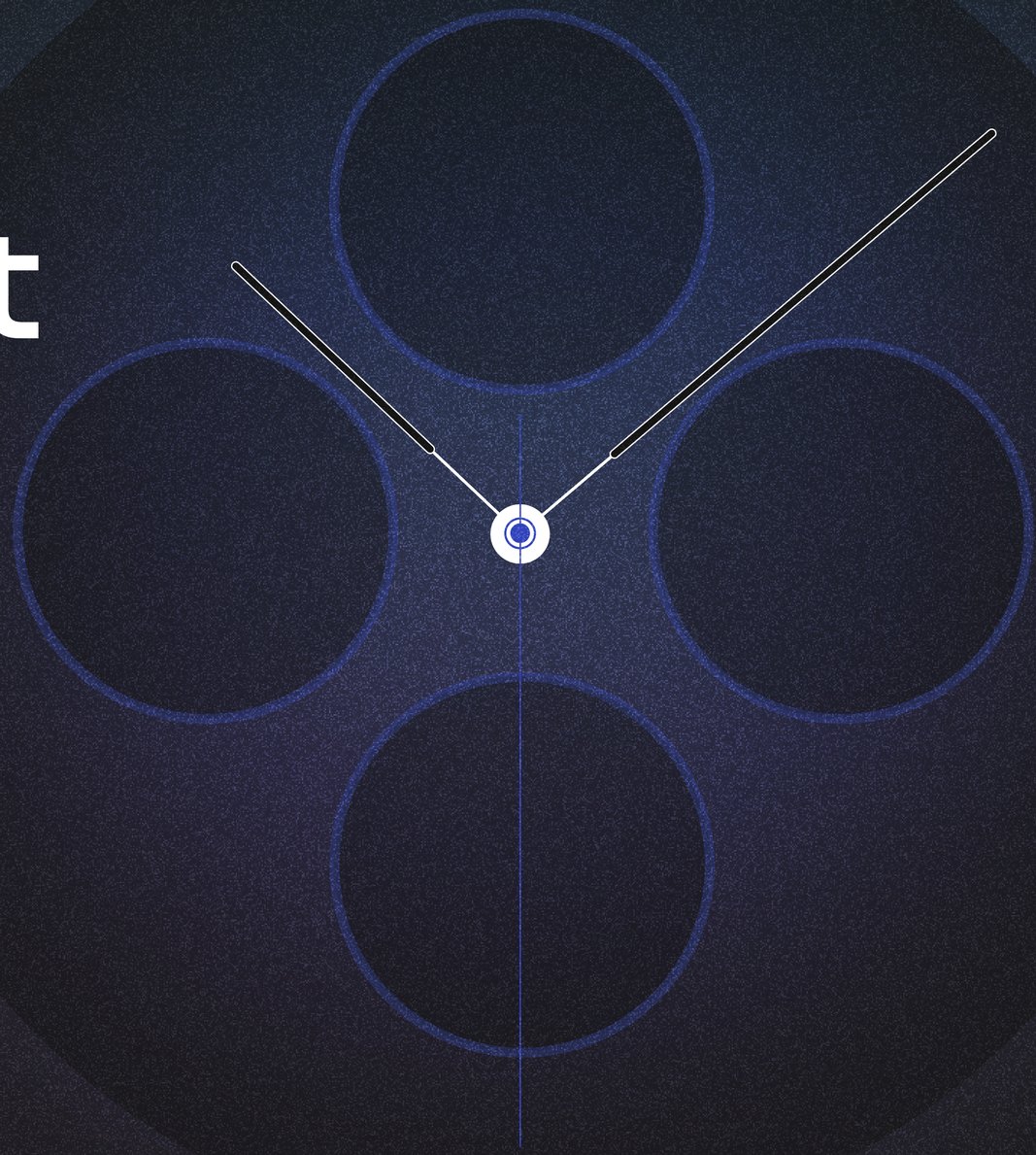
Design Requirements

Engaging and realistic cues	Environmental cues should be achievable, engaging, and realistic for users.
Nonjudgemental measurement	Success should be measured in a nonjudgemental manner to encourage users.
360 degree cues	Cues should range in directionality encouraging the user to explore the world.
Non-stressful environment	All time related environmental cues should be relaxing and calming for the user.
Incentives for completing milestones	The world should be reward based to encourage participation and engagement.

Design Principles

Design with users, not just for them	No one on our team is neurodivergent so we want to bring in our users to the design process.
Time is a unique experience	Time is a personal for everyone, the tool we design needs to honor this custom experience.
Always question “Why VR”	VR is not a perfect medium, we need to be mindful of its issues eg: overstimulation
Accommodate for different types of time	Allow users to define different types of time in their lives and customize to their preferences.
Solve one problem, expand to many	A tool designed for neurodivergent adults can be applicable to neurotypical adults as well.

Final Product



An immersive solution

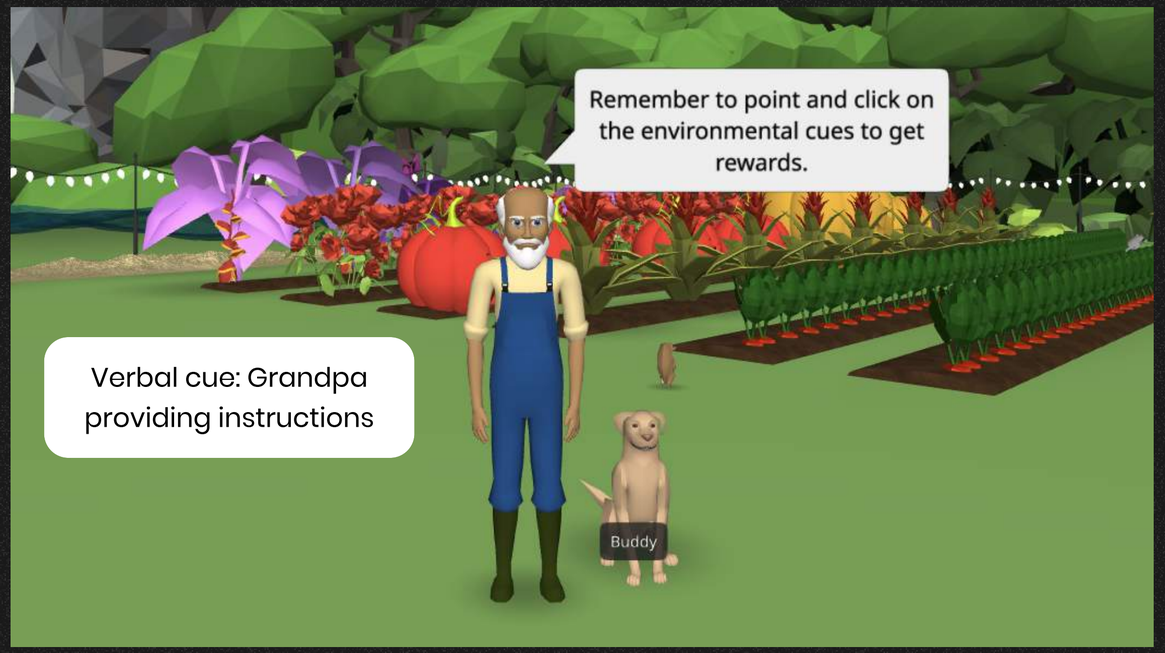
An open world VR game that rewards people for noticing time related changes in their environment



Welcome screen with tutorial



Relaxing environment for users to practice



Verbal guides in the environment for cue identification



Non Verbal guides in the environment for cue identification



Rainbow fading away around dusk

Scuba divers coming out of the water as the sun is setting



Sun setting as the moon becomes visible

Parrots flying home around evening



Crabs coming out around nighttime



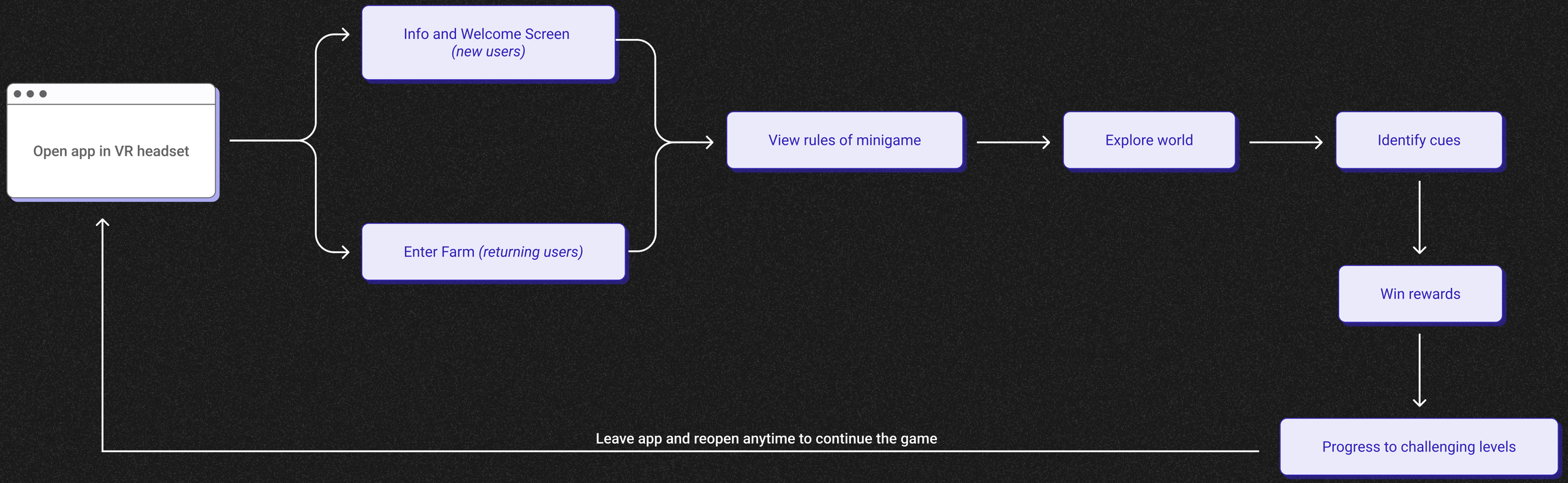
Fireflies and owls to indicate nighttime

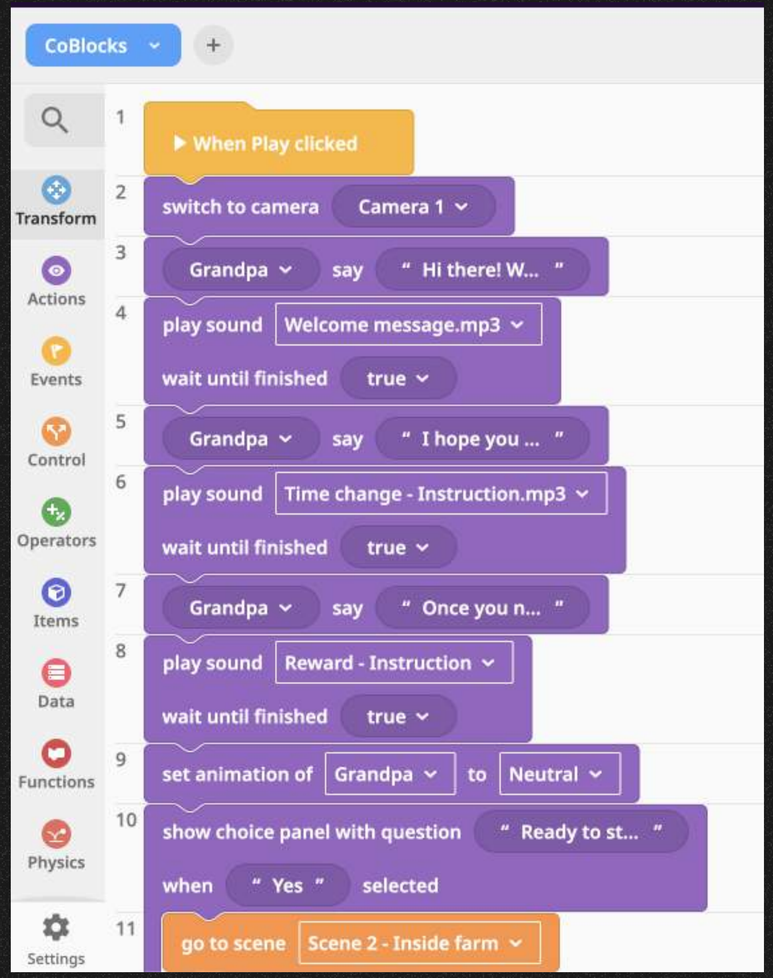
Multiple time related environment changes to interact with



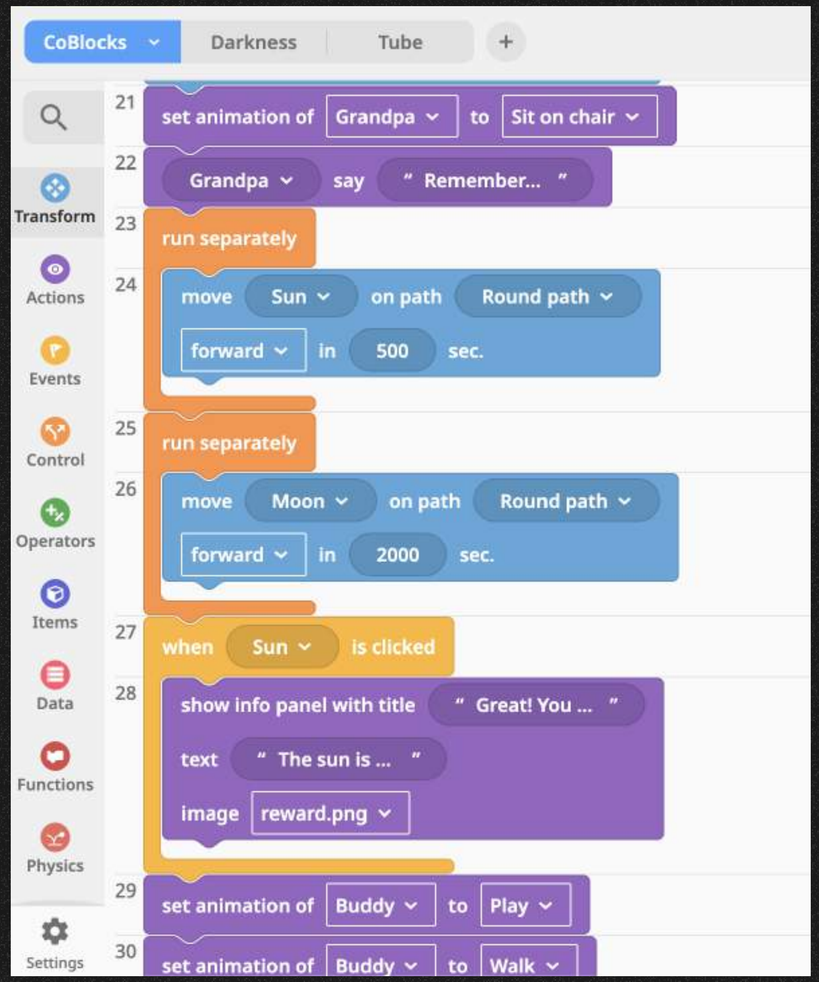
Rewards for identifying cues

Overall User Flow

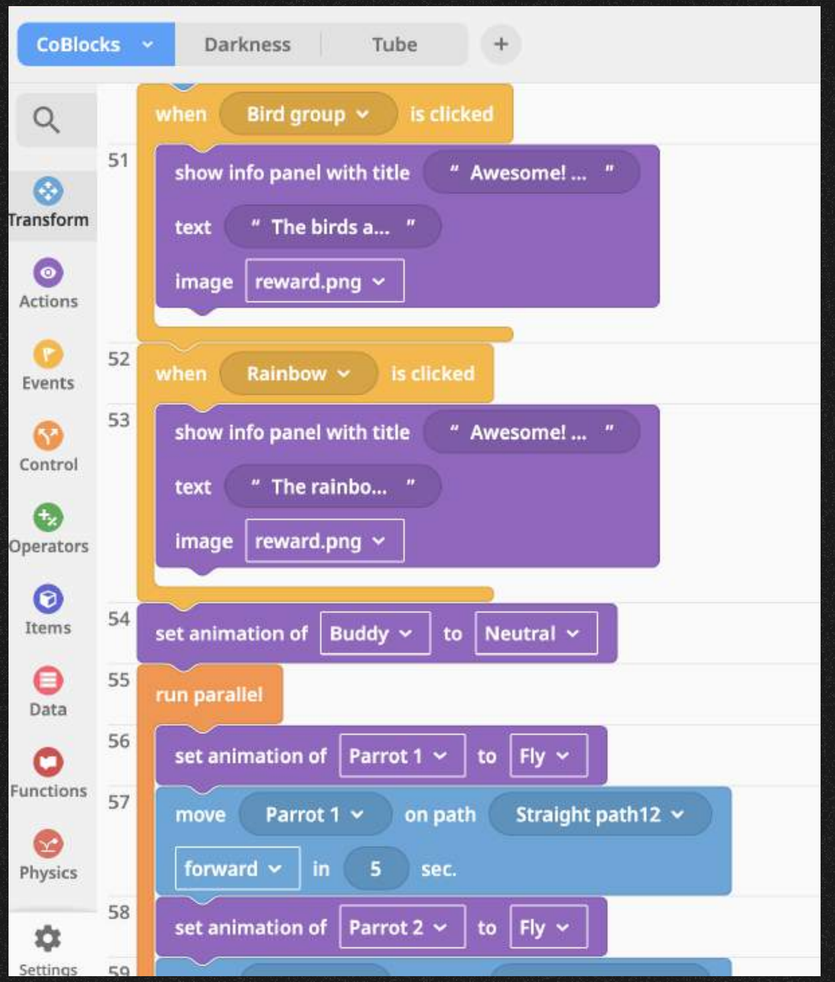




CoBlocks: Intro Scene



CoBlocks: Cues



CoBlocks: Rewards



Co-Spaces: Camera and Path Setup

How this environment helps with time perception

Improved spatial awareness

The tool trains people to tell time by observing the changing environment, this would help them build that muscle to implement in the physical world.

Building positive association with time

The environment and time related cues are calming and would help users practice time estimation in a non-stressful setting helping them form a positive association with time

Reduced cognitive overhead

The tool reduces cognitive overhead for users by gamifying spatial awareness related to time using levels and rewards.

How time works in this environment

No clocks, need to feel time with the changing environment

Changes in the environment get progressively more minor to train them to pick up on more subtle cues

24 hours maps 1:1 to the physical world, can go back in the virtual space anytime. Tasks adjust depending on time of day.

Why VR

1.

Controlled environment where changes in the surroundings get progressively minor

2.

Controlled environment to prevent stressors

3.

VR is an effective training tool for mindfulness, awareness, and noting techniques.

Limitations & Challenges



Limitations

Time warping in Virtual Reality	We're designing a tool to help with time perception but VR might warp time for users.
VR can be overstimulating	Too much time in VR can be overstimulating, our tool needs to be effective in shorter durations.
Effectiveness is a hypothesis	The effectiveness of our tool to help with time perception hasn't been tested and validated.
Tested with limited sample size	We've tested prototypes with a small sample size, this might not represent the whole audience
Untested sense of self in VR	We're unsure how connected users feel to their character in VR and whether it impacts their ability to apply learnings to the physical world.

Challenges

Problem finding and definition	Spent significant time finding the right problem to solve, leaving less time for prototyping.
Designing for neurodivergence	We were very mindful about not creating a product that causes more issues for our users
Finding non-stressful cues	Finding non-stressful time related cues was a challenge, we didn't want to trigger our users.
Managing VR controllers	VR controllers are confusing for new users, we struggled with finding ideal triggers for actions
Incremental levels and difficulty	Since users started at a different skill level we struggled making cues challenging yet achievable

Future Work



Up Next

Gamification, levels, and rewards	We hope to build out the level based reward system to experiment with gamification.
Time related cue refinement	Some cues are difficult to select using the controllers, this needs further experimentation
Task based interactions	For higher difficulty levels we want to introduce a task along with identifying environment cues
Accomodate VR constraints	Users can't spend a ton of time in VR, we need to ensure our tool is effective in shorter durations
Measuring effectiveness	We need to build out tests to ensure this tool is applicable to our users day to day life.

Reflection



Contribution to HCD field

A gap in the market

Our research suggests that while there are projects that support neurodivergent children, there is a gap in the amount of tools available for adults. Our project attempted to fill that gap.

Serious Games

Our VR game helps people practice time perception in a low-stress environment where they don't feel like they're being tested. We believe that this would be a valuable contribution to the Serious Games initiative.

Following the users

In industry it's challenging to stay true to the users needs due to business constraints. In this project, we tied every little detail back to a user need and insight. CoDesigning with our users helped us achieve this.

Implications, dangers, opportunities

1.

We haven't tested the product with children and we're worried that since its designed like a game children might be drawn to it more than adults.

2.

Level based games are notorious for having users spend large amounts of time trying to get to the end. We want to be careful encouraging long durations.

3.

If a user only logs in at the same point in the day everyday, they won't be getting exposed to cues for other times of the day.

4.

We could use our learnings to apply time based environmental changes in other VR interfaces to inform people of the amount of time passed.

5.

While research suggests that people on the neurodivergent spectrum feel comfortable in VR environments we'd need through testing to validate it.

6.

In terms of ethics, we believe that we could have come up with a solution that doesn't use technology at all with similar results.

What we'd do differently

Reflect on biases early in the process	We had to retroactively fix issues created because of biases we had while prototyping.
Test with diverse audiences	Test the product with people from a wide range of backgrounds to ensure inclusive changes.
Evaluate risks associated to time	Test the effect of our product when it comes to time - is it creating a new problem or solving one?
Long term effects of VR on time	Reduce the amount of time people spend in our product to limit VR strain
Experiment with non-digital solutions	We'd love to build non-digital prototypes to see if we can get the same results without technology

Personal Growth

During this project, I gained valuable insights from our co-design activity, realizing the individuality of human perception of time. Our inclusive solution aimed to cater to all, especially people with neurodivergence. Additionally, teamwork was pivotal as we navigated uncertainty and collaborated tirelessly with participants to develop a functional VR prototype.

Parag



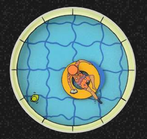
My learnings came from the 5 senses activity & testing we conducted with our users. I was surprised to see how people with ADHD and Autism use their senses to perceive time. We created our VR concept around the five senses activity and I learned that even though tasks seem easier in the real world users might have difficulty in VR, so we scaled down the final concept based on that.

Haasini



Instead of designing what the participant TELL, I learned to build what they NEED behind the WHY of their magic wand answer for the 5 senses activity. Design is a non-linear process, so I co-design with iterative VR prototypes, informed by insights from individuals with ADHD and Autism. I learned to apply the double-diamond design thinking framework to my approach.

Yifan



My biggest learnings occurred through our co-design process. From setting them up, running them, and synthesizing findings, I came out on the other side as a more thoughtful designer. Designing with your users and not simply for them from afar is a game changer when it comes to creating effective solutions. I'm excited to apply these learnings to industry.

Gresshaa



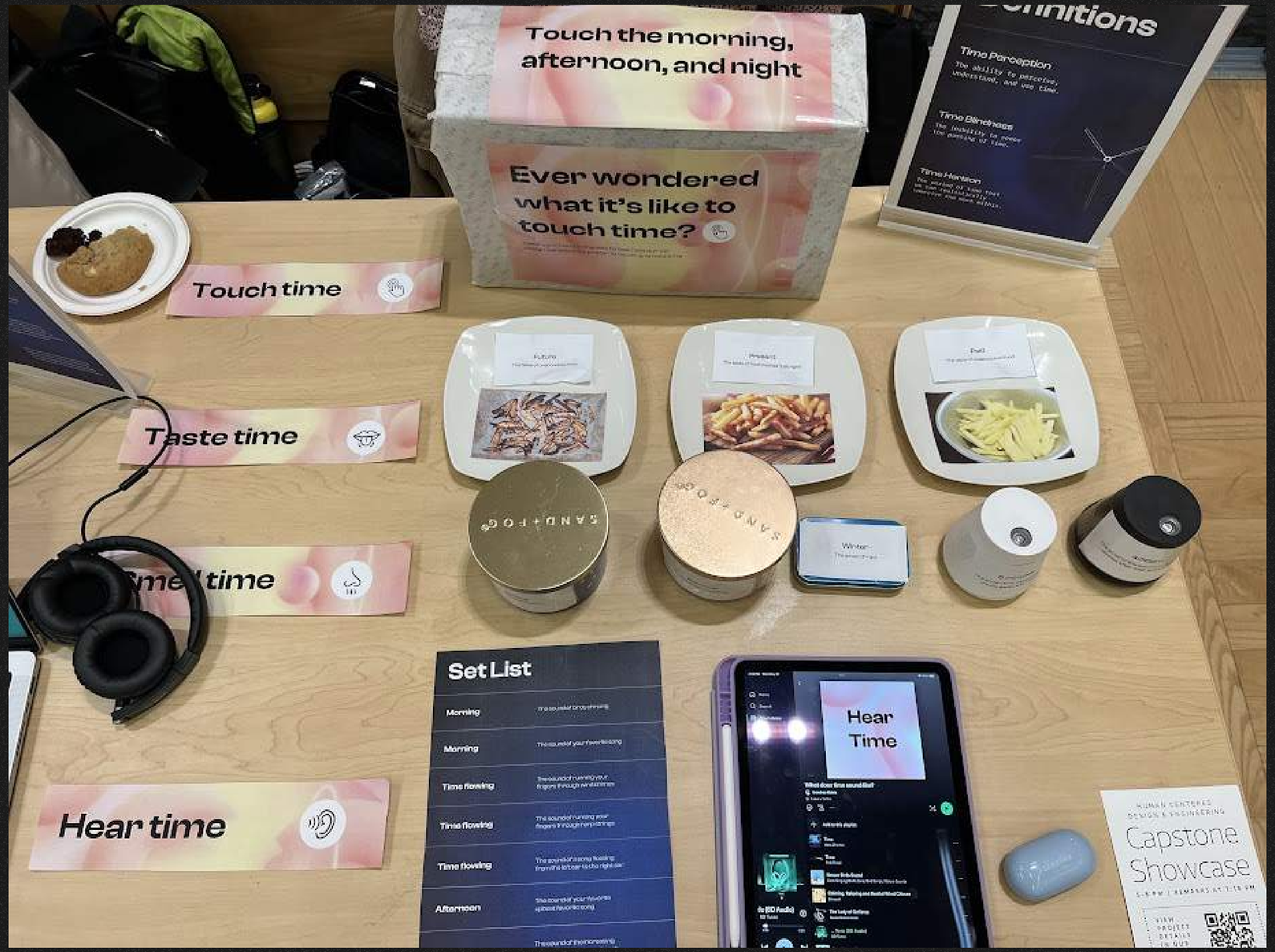




Capstone participants engaging with our prototype



Poster



Interactive Activity: Bringing the 5 senses co-design to life

Touch time





Taste time



Smell time



Hear time

Zooming into touching, tasting, smelling, and hearing time



Capstone Showcase Spring 2023 | Team Fantastic Four

Appendix



References	Docs: https://tinyurl.com/VR-Fantastic-Appendix
Project Assets	Docs: https://tinyurl.com/VR-Fantastic-Appendix
Final Prototype	CoSpaces: https://edu.cospaces.io/BFR-HXZ
Video	YouTube: https://youtu.be/TnwVGd8I0B8
Poster	Figma: https://tinyurl.com/VR-Fantastic-Poster